

Decision Journal: Week 7 Model Choice

Context

The ED Board asked whether a more complex model provides enough benefit to justify its added training cost, the ongoing work required to monitor and update it, and its reduced clarity.

Alternatives Considered

The following models were tested:

- Logistic Regression
- Tuned Random Forest
- Gradient Boosting
- Small MLP neural network
- Stacked Ensemble combining Logistic Regression and the tuned Random Forest

Each model was also evaluated using different feature combinations: with and without age, with and without MAP, and with and without engineered features. The best-performing model was Logistic Regression using age, but excluding MAP and the engineered features.

Decision

I selected Logistic Regression as the primary model for Phase 3 and will use the tuned Random Forest as a comparison model.

Reasoning

Logistic Regression achieved the highest macro-F1 score at **0.508**, compared with **0.481** for the tuned Random Forest. It also trained in **6.2 seconds**, compared with **241.7 seconds** (4.03 minutes) for the tuned Random Forest and **720.2 seconds** (12.00 minutes) for the stacked ensemble.

The stacked ensemble achieved the highest ESI 1 recall at **0.750**, but its macro-F1 was lower at **0.448**, and its ESI 2 recall was only **0.550**. This means it identified more ESI 1 patients but performed less consistently across the five ESI levels.

Logistic Regression also produced results that were easier to explain, review, and update. The more complex models did not improve overall performance enough to justify their longer training times and lower clarity.

Remaining Questions

Further analysis is needed to determine:

- How safely Logistic Regression and the tuned Random Forest handle ESI 1 and ESI 2 patients, particularly undertriage.
- Whether performance remains stable across different data splits, future patients, patient groups, and hospitals.